

2022 European guideline for the management of balanoposthitis

Abstract

Background: This guideline updates the 2014 edition on the management of balanoposthitis. Balanoposthitis refers to inflammation of the glans penis and prepuce, with diverse causes including infections, dermatoses, and premalignant conditions.

Objective: To improve recognition of symptoms, signs, and complications of penile skin conditions and to provide evidence-based recommendations on diagnosis and treatment for selected common or significant conditions.

Methods: The previous guideline was revised based on a literature review, prioritizing evidence from randomized controlled trials and systematic reviews.

Results: Key updates include: - Clearer management guidance for **Group A streptococcal infections** - Updated recommendations for **lichen sclerosus**, including circumcision and supportive care to reduce recurrence of genital herpes and warts - Additional treatment options for **Zoon's balanitis (plasma cell balanitis)** - Inclusion of **calcineurin inhibitors** in therapeutic strategies - Emphasis on the **risk of premalignancy** - Change in terminology from "premalignant conditions" to **Penile Intraepithelial Neoplasia (PeIN)**

Conclusion: Balanoposthitis has multiple etiologies. High-quality evidence specific to penile skin diseases remains limited for many conditions.

Introduction

This guideline aims to assist clinicians in recognizing the symptoms, signs, and complications of penile skin disorders that may present across specialties such

as dermatology, sexual health, and urology. It provides recommendations on diagnosis and management but is not a comprehensive review of all forms of balanoposthitis.

The guideline focuses on a selected group of conditions considered common or clinically significant, particularly those managed in dermato-venereology or sexual health settings, either independently or in collaboration with other specialists or primary care providers.

Conditions covered:

- Candidal balanoposthitis
- Anaerobic infection
- Aerobic infection
- Lichen sclerosus
- Lichen planus
- Zoon's balanitis (plasma cell balanitis)
- Psoriasis and circinate balanitis
- Seborrheic dermatitis
- Irritant/allergic eczema
- Fixed drug eruptions
- Pre-malignancy or suspected malignancy

The guideline applies primarily to individuals aged 16 years and older.

Aetiology

Balanitis refers to inflammation of the glans penis; **posthitis** refers to inflammation of the prepuce. When both are involved, the term **balanoposthitis** is used. It encompasses a range of unrelated conditions with similar clinical presentations affecting this anatomical site (see Table 1).

Balanitis is rare after circumcision, and preputial dysfunction often contributes to its development. Infections—particularly candidal—are frequently secondary to underlying inflammatory dermatoses.

Other less common dermatoses may also cause balanoposthitis but are not detailed in this guideline.

General Management of the Patient with Balanoposthitis

Clinical Features

Symptoms and signs vary by underlying cause. Specific features of individual conditions are described in their respective management sections.

Diagnosis

- Balanitis and balanoposthitis are descriptive terms encompassing various unrelated conditions. Clinical appearance may be suggestive but is never pathognomonic.
- **Biopsy** is sometimes required to exclude pre-malignant or malignant disease, especially in persistent, treatment-resistant, or atypical cases.

Recommended diagnostic investigations in uncertain cases:

- Detailed **sexual history**, including assessment of sexual risk behaviors
- **Full STI screening** (including HIV), guided by history and clinical presentation, in accordance with current guidelines
 - **HSV nucleic acid amplification test (NAAT)** if ulceration is present